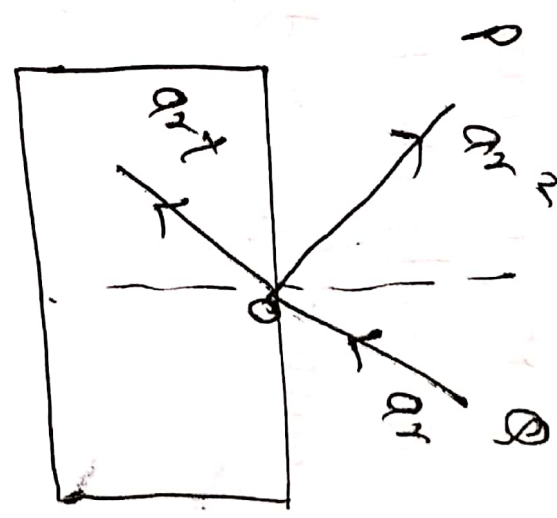
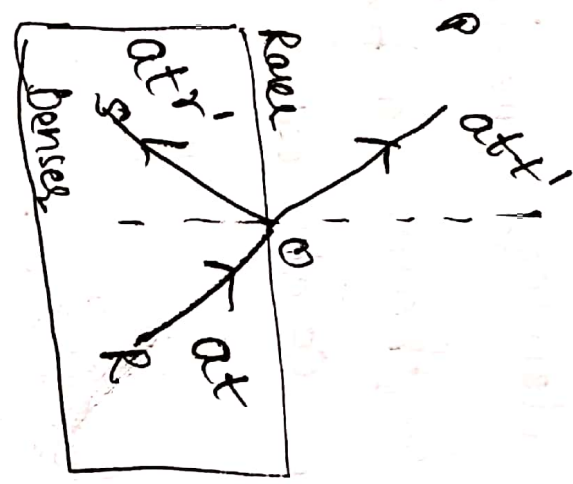
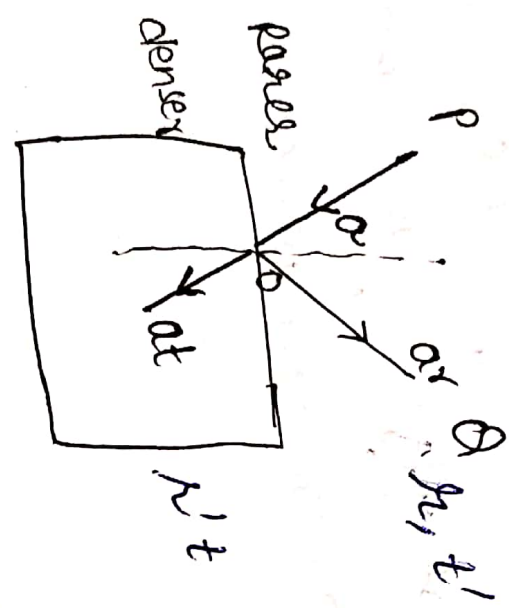


$$\frac{600}{100} = 0.6$$

$$\mu = 1.6$$

Change of phase by Reflection and Division of Amplitude

Stokes laws:-



$r = \frac{\text{Amplitude of reflected wave}}{\text{Amplitude of incident wave}}$

$t = \frac{\text{Amplitude of transmitted wave}}{\text{Amplitude of incident wave}}$

another:

$$\Rightarrow ar^2 + at t' = a$$
$$\Rightarrow t t' = r^2$$

and

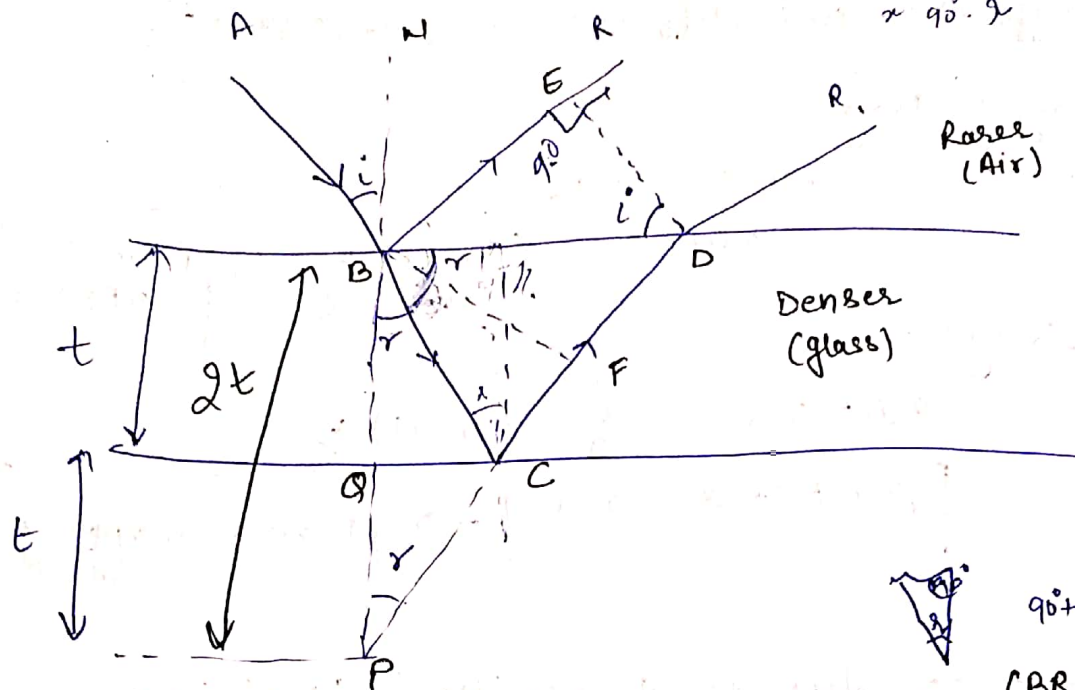
$$a r t + a r' t = 0$$

$$at(r + r') = 0$$
$$\Rightarrow r' = -r$$

$r' = -r$ $\rightarrow$ -ve sign indicates opposite dir., $\Rightarrow$ phase change of $\pi \rightarrow$ path difference of $\lambda/2$.

Principle of Reversibility $\rightarrow$ If there is no absorption, then Reflected or Refracted light rays can retrace their paths in reverse direction

$\Rightarrow$ Stokes law states that if waves are reflected at a rare to denser medium interface (eg. air-glass) the reflected waves have a phase diff. of π or path path diff. of $\lambda/2$ in comparison to incident waves.



The optical path difference b/w Reflected rays (BR and DR₁) is given by

$$\Delta = \text{Path (BC + CD) in medium} - BE \text{ in air}$$

$$= \mu (BC + CD) - BE$$

$$\mu = \frac{\sin i}{\sin r} = \frac{\frac{BE}{BD}}{\frac{FD}{BD}} = \frac{BE}{FD} \Rightarrow \boxed{BE = \mu (FD)}$$

$$\Rightarrow \Delta = \mu (BC + CD) - \mu (FD) = \mu (BC + CD - FD)$$

$$= \mu (BC + CF + FD - FD)$$

$$\boxed{BC = PC}$$

$$\Delta = \mu (PC + CF) = \mu (PF)$$

from $\triangle BPF$,

$$\cos r = \frac{PF}{BP}$$

$$\boxed{PF = BP \cos r = 2t \cos r}$$

$$\Delta = \mu \times 2t \cos r = 2\mu t \cos r$$

So effective path difference = $2\mu t \cos r \pm \frac{\lambda}{2}$
 for bright fringe

$$2\mu t \cos r \pm \frac{\lambda}{2} = n\lambda$$

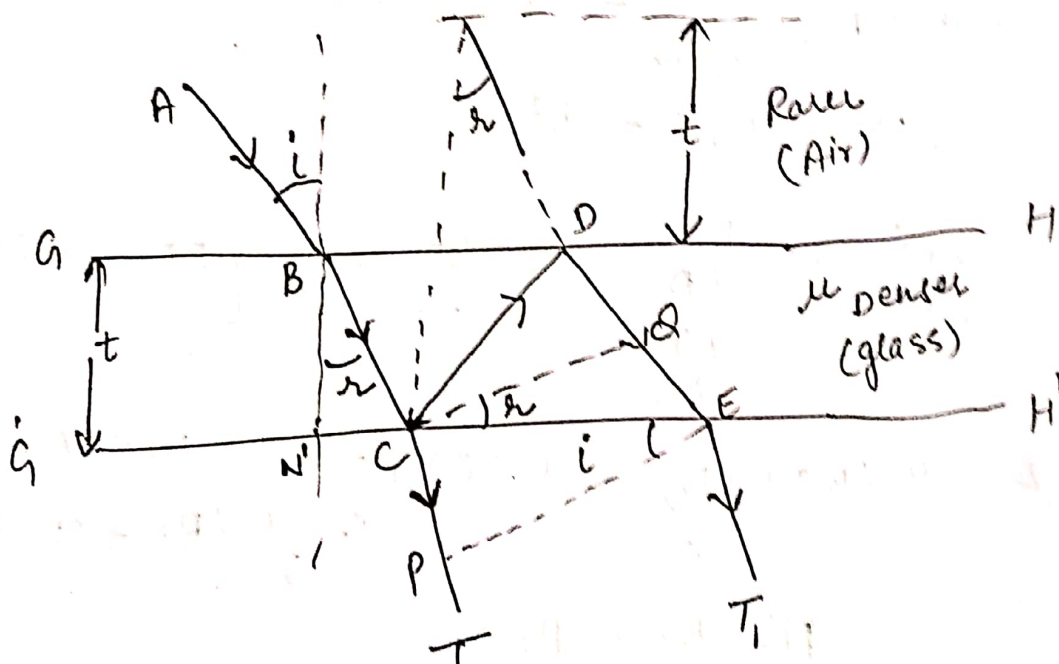
$$2\mu t \cos r = (2n \pm 1) \frac{\lambda}{2} \rightarrow \boxed{\text{Film will appear bright}}$$

for dark fringe

$$2\mu t \cos r \pm \frac{\lambda}{2} = (2n \pm 1) \frac{\lambda}{2}$$

$$\boxed{2\mu t \cos r = n\lambda} \rightarrow \boxed{\text{Film will appear dark}}$$

Interference due to transmitted light :-



$$\Delta = \mu(CD + DE) - CP$$

$$\mu = \frac{\sin i}{\sin r} = \frac{CP/CE}{QE/CE} = \frac{CP}{QE}$$